

CSE 300

LaTeX, Beamer & TikZ

Complete syntax map, creative patterns, and quiz practice

Every distinct command family in Tutorial I, Tutorial II, the two-column demo, the Beamer source, the TikZ source, and the supplied practice packs is classified here. Repetition is compressed, while important variations are rebuilt as case-by-case examples and quiz-sized creative extensions.

Source scope: the three course ZIP archives plus the six LaTeX practice/reference PDFs supplied for revision. Example subject matter inside those files is treated as demonstration content, not as instructions.

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1 How to study this handbook

Learn syntax as a hierarchy: **document structure** → **environments** → **commands** → **options** → **cross-references**. For each pattern, be able to (1) recognize the output, (2) write a minimal version, (3) repair a broken version, and (4) explain the package dependency. Gold boxes are especially suitable for short questions.

Part I

LaTeX Foundations

2 Document anatomy and compilation

```
\documentclass[11pt,a4paper]{article} % class + global options
\usepackage[margin=1in]{geometry} % preamble
\usepackage{graphicx,amsmath,amssymb}
\title{Report Title}\author{Author}\date{\today}
\begin{document}
\maketitle
\tableofcontents
\section{Introduction}
Body text.
\end{document}
```

The **preamble** begins at `\documentclass` and ends before `\begin{document}`. It selects class/options, loads packages, defines colors, commands, theorem environments, and metadata. The **body** contains material that is typeset. Commands begin with a backslash; mandatory arguments use braces; optional arguments use brackets; a percent sign comments out the remainder of its source line.

Exam recall

The minimum structural skeleton is `\documentclass {...}`, `\begin{document}`, and `\end{document}`. Package order can matter: `hyperref` is normally loaded late. Compile again when the table of contents, labels, or references show stale values.

Common classes are `article`, `report`, `book`, `letter`, and `beamer`. Common options include `10pt/11pt/12pt`, `a4paper`, `twocolumn`, and sometimes `draft`. Modern UTF-8 engines often need no `inputenc`; the tutorial uses `\usepackage[utf8]{inputenc}` and `\usepackage[T1]{fontenc}` for pdfLaTeX compatibility.

2.1 Files and build cycles

Images are external files; BibTeX data live in `.bib`; auxiliary files store references. A BibTeX build normally follows *LaTeX* → *BibTeX* → *LaTeX* → *LaTeX*. A plain document with cross-references generally needs two LaTeX runs.

3 Structure, title material, and page control

Syntax	Purpose
<code>\title{...} \author{...} \date{...}</code>	Metadata; <code>\today</code> generates the compilation date.
<code>\maketitle</code>	Prints title material in the body.
<code>\section{...}</code> , <code>\subsection{...}</code> , <code>\subsubsection{...}</code>	Numbered hierarchy, normally added to the table of contents.
Starred forms such as <code>\section*{...}</code>	Unnumbered and normally omitted from the table of contents.
<code>\paragraph{Heading}</code>	Run-in or lower-level heading, class-dependent.
<code>\tableofcontents</code> , <code>\listoftables</code>	Generated lists; require recompilation.
<code>\newpage</code> , <code>\pagebreak</code> , <code>\clearpage</code>	New page; requested break; new page after flushing pending floats.
abstract environment	Structured abstract in classes that support it.

To include an unnumbered heading in the contents manually, use `\section*{Acknowledgement}` followed by `\addcontentsline{toc}{section}{Acknowledgement}`.

4 Text, fonts, color, spacing, and special characters

Built-in local commands include `\textbf` {bold}, `\textit` {italic}, `\emph` {emphasis}, `\underline` {...}, `\textrm`, `\textsf`, `\texttt`, and `\textsc`. Emphasis is semantic and changes by context: emphasized text inside already emphasized text may return upright.

Size declarations, in increasing order, are `\tiny`, `\scriptsize`, `\footnotesize`, `\small`, `\normalsize`, `\large`, `\Large`, `\LARGE`, `\huge`, and `\Huge`. Limit the scope with a group: `{\Large Local text}`. A declaration such as `\Large` continues until its group/environment ends.

With `xcolor`, the main patterns are:

```
\textcolor{red}{text}
\colorbox{yellow}{note}
\definecolor{customblue}{RGB}{0,76,153}
blue!20!white % color mixing
\usepackage[table]{xcolor} % table coloring
```

Paragraphs are separated by a blank source line. `\` forces a line break and should not be used as a substitute for paragraphs. Useful spacing commands include `\,`, `\:`, `\;`, `\!` (negative thin space), `\quad`, `\qquad`, `\hfill`, `\vspace` {...}, and `\hspace` {...}.

Reserved characters require escaping in text: `\#` `\$` `\%` `\&` `\_` `\{` `\}`; use `\textbackslash` for a printed backslash, `\~` or `\textasciitilde` for a tilde, and `\^` for a caret.

5 Lists

```
\begin{itemize}
  \item Ordinary bullet
  \item[!] Custom label
  \begin{enumerate}[label=(\alph*)]
    \item Nested numbered item
  \end{enumerate}
\end{itemize}

\begin{description}
  \item[API] Application Programming Interface
\end{description}
```

`itemize`, `enumerate`, and `description` are built in. `enumitem` supports options such as `label=(\roman*)`, `label=\arabic*`, `label=(\alpha*)`, `start=3`, `leftmargin=*`, and spacing control. Lists can be mixed and nested.

6 Labels, references, links, and footnotes

`\label {key}` records the current numbered object. Use `\ref {key}` for its number and `\pageref {key}` for its page. Prefixes such as `fig:`, `tab:`, `eq:`, `sec:`, `alg:`, and `lem:` improve maintainability. Put a figure/table label *after* its caption so it captures the correct counter.

With `hyperref`, use `\href {https://example.com}{display text}` or `\url {https://example.com}`. Configure links with `\hypersetup {colorlinks=true,linkcolor=blue,citecolor=blue,urlcolor=teal}` or use the `hidelinks` option.

Footnotes use `main text\footnote {note}`. The tutorial demonstrates a repeated marker with a labelled first footnote and `\footnotemark [\ref {fn:key}]`; use this sparingly because repeated-footnote machinery can be fragile.

Part II

Mathematics and Formal Material

7 Math modes and core constructions

Every mathematical symbol belongs in math mode. Use $\dots$ or $\backslash(\dots\backslash)$ inline; $\backslash[\dots\backslash]$ for unnumbered display; **equation** for numbered display; and **equation*** for unnumbered display. The tutorials show $\$ \$ \dots \$ \$$, but $\backslash[\dots\backslash]$ is the preferred LaTeX form.

Construction	Syntax
Sub/superscript	x_i , $x_{\{i,j\}}$, x^2 , $x_i^{\{k\}}$
Fraction/root	$\frac{a}{b}$, $\sqrt{x}$, $\sqrt[n]{x}$
Text in math	$x > 0$ $\text{if } n \in S$ (requires amsmath)
Calligraphic/blackboard/script	$\mathcal{A}$, $\mathbb{R}$ (amssymb), $\mathscr{P}$ (mathrsfs)
Decorations	$\hat{x}$, $\widehat{xyz}$, $\bar{x}$, $\overline{AB}$, $\tilde{x}$, $\vec{x}$, $\dot{x}$, $\ddot{x}$, f' , f''
Braces	$\underbrace{a+\cdots+a}_n \text{ terms}$, $\overbrace{\dots}^{\dots}$
Operators	$\sin$, $\cos$, $\log$, $\ln$, $\exp$, $\lim$, $\min$, $\max$, $\det$, $\gcd$
Custom operator	$\DeclareMathOperator{\argmax}{arg\,max}$ in the preamble

7.1 High-yield symbols

Greek letters are case-sensitive. High-yield forms are:

```
\alpha \beta \gamma \delta \epsilon \varepsilon \theta \vartheta
\lambda \mu \pi \rho \sigma \phi \varphi \omega
\Gamma \Delta \Theta \Lambda \Pi \Sigma \Phi \Psi \Omega
```

Sets and logic:

```
\in \notin \subset \subseteq \supseteq \cup \cap \setminus
\varnothing \forall \exists \neg \vee \wedge \implies \iff
\models \vdash \mathbb{N} \mathbb{Z} \mathbb{Q} \mathbb{R} \mathbb{C}
```

Relations and operations:

```
\leq \geq \neq \approx \equiv \sim \simeq \cong \propto
\prec \preceq \parallel \mid \bmod \pm \mp \times \div \cdot
\circ \oplus \otimes \partial \nabla \ell \hbar
```

Arrows:

```
\to \rightarrow \leftarrow \Rightarrow \Longrightarrow
\leftrightharrow \mapsto \longmapsto \uparrow \downarrow \leadsto
```

Variable-sized operators: $\sum_{i=1}^n$, $\prod$, $\int_a^b f(x) dx$, $\iint$, $\iiint$, $\oint$, $\bigcup$, and $\bigcap$. Limits appear beside operators inline and above/below in display style.

7.2 Fences, dots, and spacing

Use matching $\left$ and $\right$ for automatic size: $\left(\frac{a}{b}\right)$, $\left\{\dots\right\}$, $\langle v, w \rangle$, $\lfloor x \rfloor$, $\lceil x \rceil$, and $\lvert v \rvert$. A one-sided delimiter uses an invisible mate: $\left. f'(x) \right|_{x=0}$. Manual sizes are $\big$, $\Big$, $\bigg$, $\Bigg$ (prefer paired forms $\Bigl$, $\Bigr$).

Use $\ldots$ for low dots in lists, $\cdots$ for centered operations, $\vdots$ vertically, and $\ddots$ diagonally. Math spacing should be semantic: $\sqrt{2}$, x , $n/\log n$, and $\int f(x) dx$.

8 Multiline equations, cases, arrays, and matrices

```
\begin{align}
f(x) &= x^2+2x+1 \label{eq:f}\\
&= (x+1)^2 \nonumber
\end{align}

|x| = \begin{cases}
x, & \text{if } x \geq 0, \\
-x, & \text{if } x < 0.
\end{cases}

A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \text{qqquad}
\det(A) = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}.
```

Use `align` for aligned numbered lines and `align*` for unnumbered lines; `&` marks alignment points and `\\` ends rows. `\nonumber` suppresses one number. `multline` breaks one long equation without alignment. `array` is a table inside math mode and requires column specifications. Matrix variants include `matrix`, `pmatrix`, `bmatrix`, `Bmatrix`, `vmatrix`, and `Vmatrix`.

Discrete-math extras from the cheat sheet include `\binom {n}{k}` (with `\dbinom` / `\tbinom`), `a\equiv b\pmod m`, `m\bmod n`, `\mod`, and `\pod`. Units should be upright text, e.g. `9.8\,\text {m}/\text {s}^2`.

9 Theorems and proofs

With `amsthm`, define environments in the preamble:

```
\newtheorem{theorem}{Theorem}
\newtheorem{lemma}[theorem]{Lemma} % shares theorem counter

\begin{lemma}[Graph Representability]\label{lem:tree}
Every tree has the stated representation.
\end{lemma}
\begin{proof}
Proof text. \qedhere
\end{proof}
```

The optional argument supplies a theorem note/title. Beamer already provides theorem-like environments. For definition/remark styles in articles, use `\theoremstyle {definition}` or `remark` before `\newtheorem`.

Part III

Tables, Figures, and Layout

10 Tables: complete pattern map

10.1 Tabular versus table

`tabular` builds the grid; `table` is a floating wrapper that supplies placement, caption, label, and list-of-tables entry.

```
\begin{table}[!htbp]
\centering
\begin{tabular}{lcr}
\toprule
Name & Score & Rank\\
\midrule
Ada & 95 & 1\\
\bottomrule
\end{tabular}
\caption{Results.}\label{tab:results}
\end{table}
```

Column specifications: `l`, `c`, `r`; `p{3cm}` fixed-width top-aligned paragraph; with `array`, `m{3cm}` vertically centered and `b{3cm}` bottom-aligned. `|` draws vertical rules; `@{...}` replaces intercolumn material/padding; `*{3}{c}` repeats a specification. `&` separates cells and `\\` ends a row.

Rules: `\hline` spans all columns; `\cline {2-3}` spans selected columns. The professional `booktabs` set is `\toprule`, `\midrule`, `\bottomrule`, and `\cmidrule (lr){1-2}`; avoid vertical rules with `booktabs`.

10.2 Merging, color, widths, and long tables

```
\multicolumn{2}{c}{Merged heading}
\multirow{2}{*}{Shared row} & A & B\\
& C & D\\

\rowcolors{2}{gray!10}{white}
\cellcolor{green!15}Correct
```

`\multicolumn {n}{spec}{text}` is built in; `\multirow {rows}{width}{text}` needs `multirow`. Row/cell color needs `xcolor`'s `table` option. For automatically stretched columns use `tabularx`: `\begin {tabularx}{\linewidth}{1Xr}`. For page-breaking tables use `longtable`; it is not placed inside `table`. For a table too wide, first redesign or use paragraph columns; as a last resort use `\resizebox {\linewidth }{!}{...}` from `graphicx`.

Common trap

A five-column table whose first column is left aligned, last is right aligned, middle three centered, and has no vertical lines uses `\begin{tabular}{lcccr}`, not square brackets and not vertical bars. The argument is a mandatory brace argument.

11 Figures and image variations

11.1 Single image and options

```
\usepackage{graphicx}
\graphicspath{{images/}}
\begin{figure}[!htbp]
  \centering
  \includegraphics[width=.65\linewidth,
                  height=5cm,keepaspectratio,
                  angle=0]{plot.pdf}
  \caption{A result.}\label{fig:result}
\end{figure}
```

`width` and `height` set dimensions; `keepaspectratio` prevents distortion when both are given; `scale` applies a factor; `angle` rotates; `trim=left bottom right top,clip` crops. Vector PDF is preferred for diagrams/plots; PNG/JPEG suit raster content.

11.2 Float placement

Code	Meaning	Reality
<code>h</code>	here	Try near the source location.
<code>t/b</code>	top/bottom	Permit top or bottom.
<code>p</code>	float page	Permit a page of floats.
<code>!</code>	relax restrictions	Does not mean exact placement.
<code>H</code>	exactly here	Requires <code>float</code> ; use sparingly.

Order inside a placement list does not create an execution sequence; LaTeX applies its float algorithm. `[!htbp]` is a flexible default. Capital T or B are not ordinary float options.

11.3 Two images without and with subcaptions

Two independent images can be placed on one line with widths below half the line and `\hfill`. If they need individual subcaptions/references, use `subcaption`:

```
\begin{figure}[ht]
\centering
\begin{subfigure}[t]{.48\linewidth}
  \centering\includegraphics[width=\linewidth]{a.png}
  \caption{First}\label{fig:a}
\end{subfigure}\hfill
\begin{subfigure}[t]{.48\linewidth}
  \centering\includegraphics[width=\linewidth]{b.png}
  \caption{Second}\label{fig:b}
\end{subfigure}
\caption{Comparison.}\label{fig:both}
\end{figure}
```

Use `\ref{fig:both}` for the whole figure and `\subref{fig:a}` (or an ordinary `\ref`) for a part. A blank source line starts a new paragraph and can force the next subfigure to a new row. A 2-by-2 grid repeats the pair after an explicit `\par` or vertical separation; each slot remains slightly below half width.

11.4 Figure beside text and two-column figures

For text wrapping, `wrapfig` provides `\begin{wrapfigure}{r}{.35\linewidth}`. The first argument is side (`l/r`); the second is width. Keep enough following text and avoid overly wide wrapped figures.

For exact side-by-side text and figure, a robust creative option is two `minipages`:

```
\begin{minipage}[t]{.55\linewidth}Text...\end{minipage}\hfill
\begin{minipage}[t]{.40\linewidth}
\centering\includegraphics[width=\linewidth]{a.png}
\end{minipage}
```

In a `twocolumn` article, ordinary `figure` uses one column and `\linewidth` is that column's width. `figure*` spans both columns and normally floats to a page top; use `\textwidth` or subfigure widths based on it.

12 Two-column and multi-column text

Document-wide columns use:

```
\documentclass[twocolumn]{article}
```

Local balanced columns use `multicol`:

```
\begin{multicols}{2} ... \end{multicols}
```

These are different mechanisms; double-column floats belong to true two-column output. `\linewidth` means the current line width; `\textwidth` is the full text block, so the distinction matters inside columns and minipages.

Part IV

Bibliography, Algorithms, and Code

13 BibTeX and natbib

```
@inproceedings{vaswani2017attention,
  author={Ashish Vaswani and others},
  title={Attention Is All You Need},
  booktitle={Advances in Neural Information Processing Systems},
  year={2017}, pages={5998--6008}
}
```

Plain BibTeX uses `\cite {key}`. With `\usepackage [numbers]{natbib}`, use `\citet {key}` when the author is grammatical text, `\citep {key}` for parenthetical evidence, `\citeauthor {key}`, and `\citeyear {key}`. Multiple keys are comma-separated. `\nocite {key}` includes an uncited entry; `\nocite {*}` includes every database entry.

End the document with `\bibliographystyle {plainnat}` and `\bibliography {references}` (no `.bib` extension). Styles named in the tutorial: `plainnat`, `plain`, `abbrv`, `unsrt`, `ieeetr`, and `apalike`. `unsrt` follows first citation order; `plain` sorts alphabetically by author; `abbrv` abbreviates names; `ieeetr` is engineering-oriented.

14 Algorithms and source-code listings

The source demonstrates two alternative pseudocode ecosystems; do not casually mix their commands.

algorithm + algpseudocode:

```
\begin{algorithm}[H]
\caption{Binary Search}\label{alg:binary}
\begin{algorithmic}[1]
\Require Sorted array $A$, target $x$
\Ensure Index or $-1$
\While{$low \le high$}
  \State $mid \gets \lfloor floor(low+high)/2 \rfloor$
  \If{$A[mid] = x$} \State \Return $mid$ \EndIf
\EndWhile
\end{algorithmic}
\end{algorithm}
```

`algorithm2e` uses commands such as `\KwIn`, `\KwOut`, `\SetKwFunction`, `\SetKwProg`, `\Fn`, `\If`, `\Return`, `\tcp`, and semicolon control. Its package options in the showcase are `linesnumbered`, `ruled`, `vlined`.

For real program text, `listings` provides `lstlisting`; options include `language=Python`, `caption`, and `label`. A typical global setup is:

```
\lstset{basicstyle=\ttfamily\small,
keywordstyle=\color{blue}\bfseries,
commentstyle=\color{green!50!black},
stringstyle=\color{red},frame=single,
breaklines=true,showstringspaces=false}
```

Inline literal material uses `\verb |...|` or `\lstinline`. In Beamer, a frame containing verbatim or listings must normally be `\begin {frame}[fragile]`.

Part V

Beamer Presentations

15 Beamer anatomy, themes, and frames

```
\documentclass{beamer}
\usetheme{Madrid}\usecolortheme{seahorse}
\title{Title}\subtitle{Subtitle}\author{Author}
\institute[CSE, BUET]{Department of CSE, BUET}\date{\today}
\begin{document}
\begin{frame}\maketitle\end{frame}
\begin{frame}{Outline}\tableofcontents\end{frame}
\section{Basics}
\begin{frame}{Frame title}
Content
\end{frame}
\end{document}
```

A **frame** is a source container. A frame can generate several output **slides/pages** when overlays reveal content incrementally. A title can be the frame's argument or set with `\frametitle {...}`.

Creative extension

Useful extensions: `\begin {frame}[fragile]{Code}` for verbatim/listings; `[allowframebreaks]` for automatically split bibliographies; `\titlepage`; `\AtBeginSection []{...}` for section-outline frames; `\setbeamertemplate {navigation symbols}{}` to hide navigation icons.

16 Overlays and incremental disclosure

Syntax	Effect
<code>\pause</code>	End current overlay; following content begins on the next.
<code>\item <3-></code>	Item appears from overlay 3 onward.
<code>\item <1-2></code>	Item exists only on overlays 1 and 2.
<code>\only <2>{X}</code>	X exists only on overlay 2 and occupies no space otherwise.
<code>\uncover <2->{X}</code>	Reserves space; content becomes uncovered.
<code>\visible <2->{X}</code>	Reserves space and changes visibility, without transparent-cover behavior.
<code>\onslide <2-3> X</code>	Controls following material; braces are safer for local scope.
<code>\textbf <2>{X}</code> , <code>\textcolor <2>{orange}{X}</code>	Formatting is active only on specified overlays.
<code>\alert <2>{X}</code> , <code>\structure <2>{X}</code>	Theme-aware emphasis.
<code>\setbeamercovered {transparent}</code> makes covered material faintly visible. Overlay specifications include <code><n></code> , <code><n-></code> , <code><-n></code> , <code><m-n></code> , and comma-separated combinations.	

Common trap

An overlay is not necessarily another frame. One frame with five overlay stages can produce five PDF slides. This distinction explains questions about frame count versus slide/page count.

17 Beamer lists, columns, tables, blocks, and mathematics

Lists work as in articles, including `description`; overlay specifications can be attached to items. Beamer also supports `\begin{itemize}[<+>]` to reveal items successively.

```
\begin{columns}[T,onlytextwidth]
\begin{column}{.42\textwidth}Explanation\end{column}
\begin{column}{.55\textwidth}
\centering\includegraphics[width=\linewidth]{image.png}
\end{column}
\end{columns}
```

The source's shorter `\column{.25\textwidth}` syntax is also valid inside `columns`. Keep total widths plus gaps at or below the available width.

Tables use ordinary `tabular`. Rows can be revealed with scoped `\onslide <...>{row contents}`; careless unbraced `\onslide` may affect later material. Beamer supplies `block`, `alertblock`, and `exampleblock`. It also supplies `theorem`, `corollary`, and `proof` environments whose appearance follows the theme.

Good presentation practice: large readable type, high contrast, restrained overlays, short lines, one main message per frame, and images sized to the current column's `\linewidth`.

Part VI

TikZ and Specialized Packages

18 TikZ essentials

Load `tikz` and the libraries required by the diagram:

```
\usepackage{tikz}
\usetikzlibrary{positioning,automata,arrows.meta,calc,
                trees,mindmap,shadows}
```

Define reusable styles with `\tikzset`.

```
\tikzset{
  block/.style={draw,rounded corners,minimum width=2.6cm,
                minimum height=1cm,align=center,fill=blue!10},
  line/.style={->,thick,red}
}
\begin{tikzpicture}[node distance=1cm]
\node[block] (a) {Input};
\node[block,right=of a] (b) {Output};
\draw[line] (a) -- (b) node[midway,above] {process};
\end{tikzpicture}
```

Core drawing syntax from the tutorials:

- Coordinates and shapes include named coordinates, circles, rectangles, and closed paths:

```
\coordinate (O) at (0,0);
\draw (O) circle (3);
\draw (0,0) rectangle ++(1,1);
\draw (A)--(B)--(C)--cycle;
```

- Axes/arrows: `\draw [->,thick] (-5,0)-(5,0) node[right]{$x$}`.
- Grid/fill: `\draw [step=1cm,gray!30] (-5,-5) grid (5,5)` and `\fill (0,0) circle (4pt)`.
- Relative placement: `right=of a` needs the `positioning` library.
- Loops: `\foreach \x in {0,...,4}{...}`; nested loops can generate grids or all neural-network edges.
- Automata: node options `state`, `initial`, `accepting`; edges may use `bend left`, `loop above`, and `node{label}`.
- Trees: `\node {Root} child{node{$v_1$}} child{...}`; style levels with sibling distances.

19 The showcase package map

These packages appear in the supplied tutorial showcase. Know the purpose and recognize the top-level environment/command; deep domain syntax is less likely than dependency questions.

Package/library	Main interface	Demonstrated purpose
pgfplots	<code>axis,</code> ; <code>\addplot</code> <code>\pgfplotsset</code> <code>{compat=1.18}</code>	Plots such as e^{-x^2} inside TikZ.
pgfgantt	<code>ganttchart,</code> <code>\gantttitle</code> , <code>\ganttbar</code>	Project schedule/timeline.
circuitikz	<code>circuitikz;</code> paths such as <code>to[R,l=...], to[C,l=...]</code>	Electrical circuits.
chemfig	<code>\chemfig {...}</code>	Molecular structures.
skak	<code>\newgame</code> , <code>\mainline</code> , <code>\showboard</code>	Chess moves and boards.
musixtex	<code>music,</code> <code>\Notes</code> , <code>\qu</code> , <code>\ha</code>	Musical notation.
verse	<code>verse</code>	Poetry/lineated text.
lettrine	<code>\lettrine</code> <code>[lines=2]{M}{astering}</code>	Drop capitals.
multicol	<code>multicols</code>	Local multi-column prose.
TikZ mindmap	<code>mindmap,</code> <code>concept,</code> <code>child nodes</code>	Concept maps.
tikz-qtree	<code>\Tree [...]</code>	Linguistic syntax trees (commented example).
tcolorbox	<code>tcolorbox</code>	Styled content boxes.
lipsum	<code>\lipsum [1-6]</code>	Placeholder text only.
comment	<code>comment</code> environment	Multi-line source comments.

20 Complete mathematics cheat-sheet inventory

The included two-page cheat sheet is broader than the worked tutorials. The earlier math sections explain the main ideas; this inventory ensures that its remaining command families are not silently omitted.

Greek variants and alphabet styles

```
\zeta \eta \iota \kappaappa \nu \xi \Xi \tau \tau \upsilon \Upsilon \chi
\varsigma \varpi \varrho
\mathcal{A} ... \mathcal{Z}
\mathscr{A} ... \mathscr{Z} % requires mathrsfs
```

Sets, logic, negation, and decorations

```
\emptyset \aleph \complement \not\subset \overline{A}
\dot{a} \ddot{a} \bar{x} \tilde{x} \hat{x} \widehat{x+y}
\vec{x} \vec{\imath} \vec{j} \boldsymbol{x}
\underbrace{x+y}_{comment} \overbrace{x+y}^{comment}
```

Named upright operators

```
\sin \cos \tan \sec \csc \cot
\sinh \cosh \tanh \coth
\arcsin \arccos \arctan
\exp \log \ln \lg \det \dim \ker \deg \gcd
\min \max \inf \sup \lim \liminf \limsup \arg
```

Relations, binary symbols, angles, and arrows

```
\ll \gg \asymp \doteq \ncong \nsim \succ \succeq
\angle \measuredangle \ast \star \ominus \odot \oslash \surd
\upharpoonright \checkmark
\nrightarrow \nleftarrow \nrightarrow \Longrightarrow
\rightsquigarrow \updownarrow
```

Large operators, fences, arrays, and manual sizes

```
\prod \iint \oint \bigcup \bigcap
\langle \rangle \lfloor \rfloor \lceil \rceil \lVert \rVert
\bigl( \Bigl[ \biggl\{ \Biggl|
\begin{array}{rcl} ... \end{array}
```

The cheat sheet also illustrates calculus limits/integrals, the gradient, normal-distribution notation, expected value and variance, set-builder notation with `\mid`, complement conventions, permutations, combinations, modulo variants, and thin/negative/display spacing. These are covered structurally in Sections 7 and 8; the exact subject formula may vary, but the syntax principles do not.

21 Further creative patterns worth knowing

21.1 Reserve invisible space and stabilize tables

`\phantom{999}` occupies the width and height of “999” without printing it; `\hphantom` reserves only horizontal space and `\vphantom` only vertical space. This is useful when blank and populated Beamer tables must retain identical cell dimensions.

```
\renewcommand{\arraystretch}{2} % double row height
\begin{tabular}{ccc}
\phantom{999} & \phantom{999} & \phantom{999} \\
\phantom{999} & \phantom{999} & \phantom{999}
\end{tabular}
```

A zero-width strut such as `\rule{0pt}{2.6ex}` can enforce row height without visible ink.

21.2 Manual bibliography, reusable commands, and multiple files

For a tiny document, a manual bibliography is possible:

```
\begin{thebibliography}{99} % width of the widest label, not a limit
\bibitem{lamport} L. Lamport, \emph{LaTeX}, 1994.
\end{thebibliography}
```

Define repeated notation with `\newcommand{\R}{\mathbb{R}}`; use an argument with `\newcommand{\vect}[1]{\boldsymbol{##1}}`. Larger documents can be split with `\input{section-file}` (textual inclusion) or `\include{chapter-file}` (page-oriented inclusion, class-dependent). Do not redefine an existing command with `\newcommand`; use `\renewcommand` only intentionally.

Part VII

Worked Casebooks

22 Mathematics casebook: choose by output shape

These cases are deliberately repetitive in structure: **target** → **complete syntax** → **why this environment** → **trap**. The goal is to make environment choice automatic.

Dependencies used here: `amsmath` supplies the display environments, `\eqref`, `\cfrac`, `\text`, and operator declarations; `amssymb` supplies symbols such as `\mathbb`.

22.1 Case M1: inline, numbered display, unnumbered display, and reference

```
Let  $f(t)$  be integrable on  $\mathbb{R}$ .
\begin{equation}
F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt.
\label{eq:fourier}
\end{equation}
The inverse relation is
\[
f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega.
\]
Equation~\eqref{eq:fourier} is the analysis relation.
```

Use `$. . . $` for mathematics inside prose, `equation` for one numbered display, and `\[. . .\]` for one unnumbered display. `\eqref` adds parentheses; `\ref` prints only the number. The thin space before `dt` separates the differential visually.

22.2 Case M2: an aligned derivation with only the last line numbered

```
\begin{align}
L[sI(s)-i(0)]+RI(s) &= V(s), \nonumber\\
sLI(s)+RI(s) &= V(s), \notag\\
I(s)(sL+R) &= V(s). \label{eq:rl-last}
\end{align}
```

Put `&` immediately before the relation to be aligned. `\nonumber` and `\notag` are equivalent. Do not place an extra `\\` after the last row; it may create unwanted vertical space or an empty numbered row.

22.3 Case M3: several lines with one number

```
\begin{equation}
\begin{aligned}
\omega_n &= \frac{1}{\sqrt{LC}}, \\
\zeta &= \frac{R}{2}\sqrt{\frac{C}{L}}.
\end{aligned}
\label{eq:second-order}
\end{equation}

\begin{equation}
\begin{gathered}
\omega_n^2 = \frac{1}{LC}, \\
2\zeta\omega_n = \frac{R}{L}.
\end{gathered}
\end{equation}
```

`aligned` is the inner, aligned box; `gathered` is the inner, centered box. The enclosing `equation` supplies one number. Never put the top-level `align` environment inside `equation`.

22.4 Case M4: gather, multiline, alignat, and subequations

```
\begin{gather}
\nabla\cdot\mathbf{E}=\rho/\varepsilon_0\\
\nabla\cdot\mathbf{B}=0
\end{gather}

\begin{multiline}
(a+b+c+d)^2=a^2+b^2+c^2+d^2\\
+2ab+2ac+2ad+2bc+2bd+2cd
\end{multiline}

\begin{alignat}{2}
x&=r\cos\theta, &\quad y&=r\sin\theta, \\
r&=\sqrt{x^2+y^2}, &\quad \theta&=\arctan(y/x).
\end{alignat}

\begin{subequations}\label{eq:maxwell-pair}
\begin{align}
\nabla\cdot\mathbf{B}&=0 \label{eq:max-b}\\
\nabla\times\mathbf{E}&=-\partial\mathbf{B}/\partial t \label{eq:max-e}
\end{align}
\end{subequations}
```

`gather` centers independent lines and takes no alignment `&`. `multiline` breaks one long formula. `alignat{2}` means two alignment pairs and inserts no automatic inter-pair gap. `subequations` creates a parent number with members such as (5a) and (5b); label both the group and individual equations.

22.5 Case M5: cases, matrices, determinants, and display style

```
\[
f(x)=\begin{cases}
x^2, & \text{if } x\geq 0, \\
-x, & \text{if } x<0,
\end{cases}
\\
A=\begin{bmatrix}a&b\end{bmatrix}c\&d\end{bmatrix},
\\
\det A=\begin{vmatrix}a&b\end{vmatrix}c\&d\end{vmatrix}.
\]
```

Inside `cases`, `&` separates value and condition; trailing spaces inside `\text {if }` matter. Environment names choose the fence: `pmatrix`, `bmatrix`, `Bmatrix`, `vmatrix`, `Vmatrix`, or bare `matrix`. In a table cell, add `\displaystyle` if sums, integrals, or limits must retain display size.

22.6 Case M6: growing delimiters and evaluation bars

```
\[
\left(\frac{ab}{right}),\quad
\left\langle\frac{ab}{right}\rangle,\quad
\left\lfloor\frac{ab}{right}\right\rfloor,\quad
\left.\frac{dv_C}{dt}\right|_{t=0}.
\]
```

```
\[
\left[-\frac{1}{s}e^{-st}\right]_{0}^{\infty}
\]
```

`\left` and `\right` must pair on the same math row. Use `\left .` or `\right .` as an invisible partner. Evaluation limits attach to the closing delimiter. If a visual bracket crosses a line break, use invisible partners on each row rather than leaving a delimiter unmatched.

22.7 Case M7: dense indices, leading indices, and spacing corrections

```
\[
\Gamma^{\mu}_{\alpha\beta},\quad
A^{\mu}_{\ \nu},\quad
\{^a{}_bX^c{}_d\},\quad
T^{\mu\nu}{}_{;\lambda},\quad
x_{i_jk},\quad
\sin^2\!\theta.
\]
```

Group multi-token indices. An empty group `{}` anchors a leading or second index group. A backslash-space in `A^{\mu}_{\ \nu}` creates a phantom horizontal stagger; `\!` pulls a symbol closer. Distinguish `\phi` /`\varphi` , `\epsilon` /`\varepsilon` , and `\rho` /`\varrho` when the target glyph matters.

22.8 Case M8: custom operators, stacked limits, and forced tags

```
% preamble
\DeclareMathOperator{\Lap}{\mathcal{L}}
\DeclareMathOperator*\{argmin\}{arg,min}

% body
\begin{equation}
\left(\frac{vc}{right})^2=\frac{GM}{rc^2}.
\tag{$\ast$}\label{eq:starred}
\end{equation}
\[\argmin_{\substack{\theta\in\Theta\\\theta\neq 0}}
\sum_{i=1}^n (y_i-f(x_i;\theta))^2
\]
```

The starred declaration places limits below in display style. `\operatorname {sgn}` is convenient for a one-off operator. `\tag` replaces the visible number without advancing the counter; `\tag *` omits parentheses.

22.9 Case M9: annotations, boxed results, units, and continued fractions

```
\[
I(s)=\underbrace{\frac{V_0}{R}}_{\text{steady state}}
-\underbrace{\frac{V_0}{R}}_{\text{transient}}.
\]
```

```
\[
\boxed{\,;i(t)=\frac{V_0}{R}(1-e^{-Rt/L}),\quad t\geq 0\,;
\]
```

For $L=10\,\text{mH}$ and $R=2\,\Omega$, $\tau=5\,\text{ms}$.

```
\[
\cfrac{1}{1+\cfrac{1}{1+x}}
\]
```

Use upright text for words and units, with a thin space between value and unit. `\cfrac` comes from `amsmath` and keeps continued fractions in display style.

Math environment decision table

Inline: $\dots$. One unnumbered display: $\displaystyle \left[\dots \right]$. One numbered line: `equation`. Many aligned numbered lines: `align`. Many aligned lines with one number: `equation+aligned`. Independent centered lines: `gather`; one number: `equation+gathered`. One long broken formula: `multline`. Multiple alignment pairs: `alignat{n}`. Lettered group: `subequations`. Piecewise definition: `cases`.

23 Table casebook: from simple grids to exam traps

The caption is conventionally above a table and the label follows the caption. The `table` float and `tabular` grid have different jobs.

Dependency map: `array` for custom columns; `booktabs` for professional rules; `multirow`; `makecell`; `tabularx`; `xcolor[table]`; `diagbox`; `caption` for `\captionof`; `threeparttable`; `siunitx`; and `longtable`. T9 also needs `graphicx`.

23.1 Case T1: bordered table with mathematics

```
\begin{table}[htbp]
\centering
\caption{Sorting complexity.}\label{tab:sort}
\begin{tabular}{|l|c|c|c|}
\hline
\textbf{Algorithm}&\textbf{Best}&\textbf{Average}&\textbf{Worst}\\
\hline
Merge sort& $O(n\log n)$ & $O(n\log n)$ & $O(n\log n)$ \\
Quick sort& $O(n\log n)$ & $O(n\log n)$ & $O(n^2)$ 
\end{tabular}
\end{table}
```

Count columns before rows: a four-column table normally has three `&` tokens per ordinary row. Use `\log`, not italic letters `log`.

23.2 Case T2: two-tier header with horizontal and vertical spans

```
\begin{tabular}{|l|c|c|c|}
\hline
\multirow{2}{*}{\textbf{Method}}&
\multicolumn{2}{c|}{\textbf{Error}}&
\multirow{2}{*}{\textbf{Order}}\\
\cline{2-3}
&\textbf{Local}&\textbf{Global}\\
\hline
Trapezoidal& $O(h^3)$ & $O(h^2)$ & $2$ \\
Simpson& $O(h^5)$ & $O(h^4)$ & $4$ 
\end{tabular}
```

The second header row begins and ends with placeholder `&` cells because those positions are occupied by the two `\multirow` s. Use `\cline{2-3}` so the partial rule does not cut through them.

23.3 Case T3: multirow families with tall fractions

```
\renewcommand{\arraystretch}{1.7}
\begin{tabular}{|c|c|c|c|}
\hline
\textbf{Family}&\textbf{Function}&\textbf{Derivative}&\textbf{Domain}\\
\hline
\multirow{3}{*}{Power}
& $x^n$ & $nx^{n-1}$ & $n \in \mathbb{R}$ \\
& $\frac{1}{x}$ & $-\frac{1}{x^2}$ & $x \neq 0$ \\
& $\sqrt{x}$ & $\frac{1}{2\sqrt{x}}$ & $x > 0$ 
\end{tabular}
```

Every row covered by a `multirow` still needs an empty placeholder. Use partial clines between family members and a full rule only when the family changes. Increase `\arraystretch` when fractions collide.

23.4 Case T4: colored grouped table and scientific notation

```
\usepackage[table,dvipsnames]{xcolor}
\begin{tabular}{|l|cc|cc|}
\hline
\rowcolor{Gray!35}
\multirow{2}{*}{\textbf{Material}}&
\multicolumn{2}{c|}{\textbf{Electrical}}&
\multicolumn{2}{c|}{\textbf{Derived}}\\
\cline{2-5}
\rowcolor{Gray!20}
&E_g&\mu_n&\sigma=1/\rho&\beta=\sqrt{\mu_n}\\
\hline
\rowcolor{Yellow!20} Silicon&1.12&1.35\times 10^{-3}&4.35\times 10^{-4}&36.7\\
\hline
\multicolumn{3}{|r|}{\cellcolor{Green!15}\textbf{Mean gap}}&
\multicolumn{2}{c|}{\cellcolor{Green!15}$1.14$ eV}\\
\hline
\end{tabular}
```

`\rowcolor` must be the first token of its row, even before `\multirow` or `\multicolumn`. Each multicolumn redeclares its own alignment and borders.

23.5 Case T5: professional rules and line breaks inside cells

```
\usepackage{booktabs,makcel}
\begin{tabular}{lcc}
\toprule
\textbf{Transform}&\makecell{\textbf{Kernel}}&\textbf{function}&
\makecell{\textbf{Typical}}&\textbf{application}&
\midrule
Laplace&e^{-st}&\makecell{transient circuits,&\control systems}&
Fourier&e^{-j\omega t}&\makecell{spectra,&\signal filtering}&
\bottomrule
\end{tabular}
```

Booktabs tables omit vertical rules and do not mix `\hline` with `\toprule` `\midrule` `\bottomrule`. A bare `\` in an ordinary cell ends the table row; `\makecell` contains an internal line break.

23.6 Case T6: fixed total width and custom paragraph columns

```
\usepackage{tabularx,array}
\newcolumntype{C}[1]{>{\centering\arraybackslash}p{#1}}
\newcolumntype{R}[1]{>{\raggedleft\arraybackslash}p{#1}}

\begin{tabularx}{.92\textwidth}{|l|C{3cm}|X|}
\hline
Name&Form&Physical interpretation\\
\hline
Dirichlet&u=g on &\partial\Omega&The value is prescribed.\\
\hline
Neumann&\partial u/\partial n=g&The flux is prescribed.\\
\hline
\end{tabularx}
```

X absorbs leftover width and wraps automatically. `>\dots` injects formatting before each cell; `\arraybackslash` restores the row-ending meaning of `\`.

23.7 Case T7: heavy mathematics inside cells

```
\renewcommand{\arraystretch}{2}
\begin{tabular}{|c|c|c|}
\hline
Object&Expression&Note\\
\hline
Matrix&\begin{bmatrix}a&b\\c&d\end{bmatrix}&2\times 2\\
\hline
Piecewise&f(x)=\begin{cases}x^2&x\ge 0\\-x&x<0\end{cases}&continuous\\
\hline
Integral&\displaystyle\int_0^\infty e^{-st}\,dt=\dfrac{1}{s}&s>0\\
\hline
Limit&\displaystyle\lim_{n\rightarrow\infty}(1+1/n)^n=e&classical\\
\hline
\end{tabular}
```

Nested environments claim their own `&` and `\`; they do not end the outer row. `\displaystyle` prevents large operators shrinking to inline size.

23.8 Case T8: three-tier header with a diagonal cell

```
\usepackage{diagbox,multirow}
\begin{tabular}{|c|c|c|c|}
\hline
\multirow{3}{*}{\diagbox[width=6em]{Layer}{Metric}}&
```

```
\multicolumn{2}{c|}{Training}&\multicolumn{2}{c|}{Inference}\\
\cline{2-5}
&\multicolumn{2}{c|}{per epoch}&\multicolumn{2}{c|}{per sample}\\
\cline{2-5}
&Time&Memory&Time&Memory\\\hline
Convolution&12.4 s&1.8 GB&3.1 ms&0.4 GB\\\hline
\end{tabular}
```

If `diagbox` is unavailable, preserve the three-row structure with a normal `multirow`; structural correctness is more important than the decorative diagonal.

23.9 Case T9: figure and independently numbered table side by side

```
\usepackage{caption}
\begin{figure}[htbp]\centering
\begin{minipage}[b]{.46\textwidth}\centering
\includegraphics[width=\linewidth]{plot.png}
\caption{Measured response.}\label{fig:response}
\end{minipage}\hfill
\begin{minipage}[b]{.46\textwidth}\centering
\captionof{table}{Extracted parameters.}\label{tab:params}
\begin{tabular}{lc}\toprule Quantity&Value\\\midrule
Time constant&$10$ ms\\ Overshoot&$4.3\%$\\\bottomrule
\end{tabular}
\end{minipage}
\end{figure}
```

Do not nest a `table` float inside a `figure` float. The `minipage` and `\captionof {table}` combination preserves independent counters. Table caption above; figure caption below.

23.10 Case T10: float footnote, table notes, decimal alignment, and long tables

```
% footnote from a float
B & 1.31\footnotemark\\
% after \end{table}
\footnotetext{Trial B was repeated.}

% cleaner table-specific notes: threeparttable
\begin{threeparttable}
\begin{tabular}{lS[table-format=4.2]}
\toprule Sample&{Mass (g)}\\\midrule A&1.20\\ B&143.75\tnote{a}\\\bottomrule
\end{tabular}
\begin{tablenotes}\footnotesize\item[a] Recalibrated value.\end{tablenotes}
\end{threeparttable}
```

The `S` column is from `siunitx`; header text in an `S` column is wrapped in braces. A multi-page `longtable` supplies its own caption and repeated-head commands (`\endfirsthead`, `\endhead`, `\endfoot`, `\endlastfoot`) and must not be placed inside `table`.

24 Figure casebook: every practical layout

Dependency map: start with `graphicx`; add subcaption for subfigures, `wrapfig` for wrapped figures, `rotating` for sidewaysfigure, `caption` for `\captionof`, and `float` only when exact H placement is truly required.

24.1 Case F1: one referenced figure

```
Figure~\ref{fig:single} shows the waveform.
\begin{figure}[htbp]
\centering
\includegraphics[width=.55\textwidth]{waveform.png}
\caption{Reference waveform.}\label{fig:single}
\end{figure}
```

Inside a float prefer the declaration `\centering` to the `center` environment, which adds extra vertical space. Use a nonbreaking space before the reference.

24.2 Case F2: two panels with one shared caption and no subcaptions

```
\begin{figure}[htbp]\centering
\begin{minipage}{.45\textwidth}\centering
\includegraphics[width=\linewidth]{input.png}
\end{minipage}\hfill
\begin{minipage}{.45\textwidth}\centering
\includegraphics[width=\linewidth]{output.png}
\end{minipage}
\caption{Input (left) and output (right).}\label{fig:pair}
\end{figure}
```

Use minipages when individual subcaptions are not required. Widths plus gaps must fit below one line width. Do not insert a blank paragraph between same-row minipages.

24.3 Case F3: three subcaptioned panels

```
\begin{figure}[htbp]\centering
\begin{subfigure}[b]{.30\textwidth}\centering
\includegraphics[width=\linewidth]{a.png}
\caption{Under-damped}\label{fig:under}
\end{subfigure}\hfill
\begin{subfigure}[b]{.30\textwidth}\centering
\includegraphics[width=\linewidth]{b.png}
\caption{Critical}\label{fig:critical}
\end{subfigure}\hfill
\begin{subfigure}[b]{.30\textwidth}\centering
\includegraphics[width=\linewidth]{c.png}
\caption{Over-damped}\label{fig:over}
\end{subfigure}
\caption{Three damping regimes.}\label{fig:damping}
\end{figure}
Panel~\subref{fig:critical} of Figure~\ref{fig:damping} settles fastest.
```

The optional [b] aligns panel bottoms. \subref prints only the letter; \ref usually prints the full subfigure number.

24.4 Case F4: grouped multi-row grid

```
\begin{figure}[htbp]\centering
{\large\bfseries Low frequency}\par\medskip
% three .30\textwidth subfigures separated by \hfill

\bigskip
{\large\bfseries High frequency}\par\medskip
% three more .30\textwidth subfigures
\caption{Six responses grouped by frequency.}\label{fig:grid}
\end{figure}
```

The heading is ordinary centered text inside the float; \par closes its paragraph. A blank line or explicit paragraph break starts the next panel row; \bigskip only adds vertical space. For four panels use about .22\textwidth ; for three use about .30\textwidth .

24.5 Case F5: wrapped figure

```
\begin{wrapfigure}{r}{.36\textwidth}
\centering
\includegraphics[width=\linewidth]{small.png}
\caption{Wrapped result.}\label{fig:wrap}
\end{wrapfigure}
Text begins immediately and continues long enough to fill the figure height...
```

Sides l/r are fixed; capital L/R permit floating. Place the environment immediately before the paragraph. Avoid lists, other floats, and placement directly after a section heading. Insufficient following text causes ugly spillover.

24.6 Case F6: rotation, framing, cropping, and forced dimensions

```
\fbox{\includegraphics[width=.3\linewidth]{a.png}}
\includegraphics[height=2.2cm,keepaspectratio]{b.png}
\includegraphics[width=.3\linewidth,angle=90]{c.png}
\includegraphics[width=.3\linewidth,trim=2cm 1cm 2cm 1cm,clip]{d.png}
\rotatebox{30}{\fbox{Rotated text}}
\resizebox{5cm}{!}{\includegraphics{e.png}}
```

Trim order is left, bottom, right, top; cropping is not visible without `clip`. Width is applied before rotation. Two explicit dimensions distort unless `keepaspectratio` is set. `sidewaysfigure` from `rotating` rotates the entire float and caption onto a landscape page.

24.7 Case F7: two-column article and a full-width figure

```
\documentclass[twocolumn]{article}
% one-column float
\begin{figure}[ht]\includegraphics[width=\linewidth]{a.png}\caption{One column.}\end{figure}
% two-column float
\begin{figure*}[t]\centering
\begin{subfigure}{.31\textwidth}...\end{subfigure}\hfill
\begin{subfigure}{.31\textwidth}...\end{subfigure}\hfill
\begin{subfigure}{.31\textwidth}...\end{subfigure}
\caption{Both columns.}
\end{figure*}
```

In normal two-column text, `\linewidth` is one column. A starred float spans `\textwidth` and normally appears at a page top.

24.8 Case F8: footnotes associated with figures

A normal body-text footnote works when discussing a figure. A footnote marker inside a float generally needs `\footnotemark` inside and `\footnotetext {...}` after the float. A footnote inside a moving caption is fragile and usually poor style; prefer a caption note, source line, or table-note mechanism.

Figure checklist

Use `\centering`; size relative to the current container; caption before label; same-row panel widths total below 1; no blank line between same-row panels; unique labels; `trim` with `clip`; compile twice; and remember that `[htbp]` is permission, while `[H]` is a command from `float`.

25 References, code, lists, footnotes, and document furniture

Dependency map: `listings`, `xcolor`; `verbatim` and `comment`; `hyperref`, followed by `cleveref` when used, plus optional `varioref`; `threeparttable`; `enumitem`[`inline`]; and `fancyhdr`, `setspace`, `multicol`.

25.1 Case X1: reusable listings styles, inline code, and external code

```
\lstdefinestyle{py}{language=Python,basicstyle=\ttfamily\small,
keywordstyle=\color{blue}\bfseries,commentstyle=\color{green!50!black},
stringstyle=\color{red},numbers=left,numberstyle=\tiny\color{gray},
frame=single,breaklines=true,showstringspaces=false,captionpos=b}
```

```
\begin{lstlisting}[style=py,caption={Euler integration.},label={lst:euler}]
# Python code here
\end{lstlisting}
Inline: \lstinline[language=Python]!y += h*f(t,y)!
External: \lstinputlisting[style=py]{euler.py}
```

Per-listing language changes are valid. `escapeinside={*}{*}` temporarily leaves verbatim mode for typeset mathematics. Avoid `minted` in a timed quiz unless shell escape and Pygments are guaranteed.

25.2 Case X2: verbatim forms and their restrictions

Use `verbatim` for a literal block, `\verb |...|` for inline literal text, and `\verb *|two spaces|` to show spaces. The character following `\verb` is an arbitrary delimiter. `\verb` cannot safely appear inside moving/command arguments such as section titles and captions; use `\texttt {\textbackslash command}` or `listings` instead. The `comment` environment can disable a source block.

25.3 Case X3: reference command matrix

Command	Typical output	Dependency
<code>\ref {k}</code>	bare number	built in
<code>\eqref {k}</code>	parenthesized equation number	amsmath
<code>\pageref {k}</code>	page number	built in
<code>\autoref {k}</code>	“Figure 2” with type	hyperref
<code>\nameref {k}</code>	referenced title text	hyperref/nameref
<code>\hyperref [k]{text}</code>	custom linked words	hyperref
<code>\cref {a,b} / \Cref {a}</code>	intelligent type names, lists, ranges	cleveref after hyperref
<code>\vref {k}</code>	contextual page phrase	varioref

Use `\crefname` to rename cleveref’s type words; for example, its three arguments may be `{figure}`, `{Fig.}`, and `{Figs.}`. Put `\phantomsection` before `\addcontentsline` when an unnumbered heading’s TOC link otherwise lands incorrectly.

25.4 Case X4: external links and PDF metadata

```
\hypersetup{colorlinks=true,linkcolor=blue,citecolor=teal,urlcolor=magenta,
pdftitle={Document title},pdfauthor={Your name},
pdfsubject={Course work},pdfkeywords={LaTeX, CSE}}
\href{https://ctan.org}{CTAN}
\url{https://ctan.org/pkg/listings}
\href{mailto:name@example.edu}{name@example.edu}
```

`\href` takes destination first, visible text second. `\url` handles line breaking and special URL characters. `hidelinks` keeps links active but removes visible boxes/colors.

25.5 Case X5: footnotes in difficult contexts

```
Body text.\footnote{Ordinary note.}

% float
Value\footnotemark
% immediately after the float
\footnotetext{Note belonging to the value.}

% moving argument: use cautiously
\section{Title\protect\footnote{Note text.}}

% symbol markers
\renewcommand{\thefootnote}{\fnsymbol{footnote}}
```

For proper notes directly below a table, `threeparttable` is cleaner: use `\tnote {a}` in a cell and `\item [a]` inside `tablenotes`.

25.6 Case X6: advanced enumitem patterns

```
\usepackage[inline]{enumitem}
\begin{enumerate}[label=(\Roman*)]
\item Main
\begin{itemize}[label=$\dagger$]
\item Nested \item[$\oplus$] One-off marker
\end{itemize}
\end{enumerate}

\begin{enumerate}[label=\arabic*.]\item First\item Second\end{enumerate}
Interruption.
\begin{enumerate}[label=\arabic*.,resume]\item Third\end{enumerate}

Inline: \begin{enumerate*}[label=(\alph*)]
\item R, \item C, and \item L
\end{enumerate*}
```

Useful keys: `start`, `resume`, `noitemsep`, `nosep`, `leftmargin`, `itemsep`, `topsep`, and `style=nextline` for descriptions. Global configuration uses `\setlist`.

25.7 Case X7: manual bibliography versus database workflows

```
Text~\cite{book}; page-specific~\cite[p.~245]{book};
multiple~\cite{paper,web}.
\begin{thebibliography}{99}
\bibitem{book} A.~Author, \textit{Book Title}, 2nd ed., 2024.
\bibitem{paper} B.~Author, ‘‘Article title,’’ \textit{Journal},
vol.~2, no.~3, pp.~10--20, 2025.
\bibitem{web} Site, ‘‘Page title.’’ [Online]. Available: \url{https://...}
\end{thebibliography}
```

The argument {99} reserves label width; it is not a maximum count. Manual entries are fastest in a lab quiz. BibTeX uses the four-pass cycle; natbib adds \citet / \citep . Modern biblatex+biber is recognition-only unless explicitly taught.

25.8 Case X8: contents depth, headers, spacing, and local columns

```
\setcounter{tocdepth}{2}\setcounter{secnumdepth}{3}
\section*{Acknowledgements}
\phantomsection\addcontentsline{toc}{section}{Acknowledgements}

\pagestyle{fancy}\fancyhf{}
\fancyhead[L]{Short title}\fancyhead[R]{\thepage}

\begin{spacing}{1.5}Looser paragraph.\end{spacing}
\begin{multicols}{2}Balanced local columns.\end{multicols}
```

Dependencies are fancyhdr, setspace, and multicols. Clear header fields before redefining them. Ordinary floats do not work reliably inside multicols; use minipages and \captionof .

26 Beamer casebook: article knowledge applied to slides

The class is beamer. The cases additionally use booktabs, graphicx, subcaption, listings, and tikz; relative node placement needs the TikZ positioning library.

26.1 Case B1: a reusable presentation skeleton

```
\documentclass[aspectratio=169]{beamer}
\usetheme{Madrid}\usecolortheme{seahorse}
\setbeamertemplate{navigationsymbols}{}
\title[Short Title]{Full Presentation Title}
\subtitle{Subtitle}\author{Author}
\institute[CSE, BUET]{Department of CSE, BUET}\date{\today}
\AtBeginSection[] {
  \begin{frame}{Section outline}
    \tableofcontents[currentsection]
  \end{frame}
\begin{document}
\begin{frame}[plain]\titlepage\end{frame}
\begin{frame}{Outline}\tableofcontents\end{frame}
\section{Methods}
\begin{frame}{Method}Content\end{frame}
\end{document}
```

aspectratio=169 selects widescreen. Optional short metadata appears in navigation areas. A plain frame removes headline/footline.

26.2 Case B2: overlay specifications and space behavior

```
\begin{frame}{Overlays}
A\pause B\pause C
\only<2>{occupies space only on slide 2}
\uncover<2->{space reserved on every slide}
\visible<3->{\visible from slide 3; space reserved}
\onslide<4->{applies to following material}
\alt<2>{shown on 2}{shown otherwise}
\temporal<3>{before}{on 3}{after}
\end{frame}
```

`\only` changes layout because absent material takes no space. `\uncover` preserves geometry and follows covered-item behavior; `\visible` preserves geometry without transparent rendering. Scope `\onslide` carefully with braces or reset it.

26.3 Case B3: staged and nested lists

```
\setbeamercovered{transparent}
\begin{frame}{Procedure}
\begin{enumerate}[<+>]
\item Setup
\item Simulate
  \begin{itemize}
    \item<.-> Monitor
    \item<+> Record
  \end{itemize}
\item<5-> Report
\end{enumerate}
\end{frame}
```

`<+>` advances the overlay counter automatically. A dot specification such as `<.->` begins on the current overlay. Use explicit numbers when exact synchronization matters.

26.4 Case B4: text-image columns and alignment

```
\begin{frame}{Architecture}
\begin{columns}[T,onlytextwidth]
\begin{column}{.42\textwidth}
\begin{block}{Idea}Short explanation.\end{block}
\end{column}
\begin{column}{.55\textwidth}
\centering\includegraphics[width=\linewidth]{architecture.pdf}
\end{column}
\end{columns}
\end{frame}
```

T top-aligns the columns; `onlytextwidth` avoids outer margins being counted twice. Widths plus gaps must fit. Use `\linewidth` inside a column.

26.5 Case B5: stable staged tables

```
\begin{frame}{Results}
\renewcommand{\arraystretch}{1.3}
\begin{tabular}{lcc}
\toprule Method&Accuracy&Time\\ \midrule
Baseline&81\%&10 s\\
\uncover<2->{Model A}&\uncover<2->{88\%}&\uncover<2->{8 s}\\
\uncover<3->{Model B}&\uncover<3->{91\%}&\uncover<3->{7 s}\\
\bottomrule
\end{tabular}
\end{frame}
```

`\uncover` keeps the table footprint stable. For a blank table on one slide and a populated table on the next, use `\phantom{999}` in cells and the same `\arraystretch`; invisible content preserves dimensions.

26.6 Case B6: images and subfigures on slides

```
\begin{frame}{Comparison}
\begin{figure}\centering
\begin{subfigure}{.47\textwidth}\includegraphics[width=\linewidth]{a.png}\caption{Before}\end{subfigure}\hfill
\begin{subfigure}{.47\textwidth}\includegraphics[width=\linewidth]{b.png}\caption{After}\end{subfigure}
\end{figure}
\end{frame}
```

Beamer does not need floating placement options; content stays in the frame. Captions consume scarce vertical space, so use them only when useful. Reveal panels with `\only` if reflow is desired, or `\uncover` if stable geometry is desired.

26.7 Case B7: blocks, theorem material, and progressive mathematics

```

\begin{frame}{Main result}
\begin{alertblock}{Problem}Naive search is exponential.\end{alertblock}
\begin{theorem}[Correctness]
The algorithm returns an optimal solution.
\end{theorem}
\begin{proof}
\only<2->{Feasibility follows from the invariant.}
\only<3->{Optimality follows from the exchange argument.}
\end{proof}
\end{frame}

```

Theme-aware blocks are preferable to manually colored rectangles. Avoid revealing individual symbols so aggressively that the audience cannot read the full equation.

26.8 Case B8: code and long content

```

\begin{frame}[fragile]{Code}
\begin{lstlisting}[language=Python]
def square(x):
    return x*x
\end{lstlisting}
\end{frame}

\begin{frame}[allowframebreaks]{References}
\bibliography{refs}
\end{frame}

```

A verbatim/listings frame is fragile. `allowframebreaks` splits long material into multiple frames and is most suitable for references; do not use it to avoid editing an overcrowded ordinary slide. `shrink` exists but readable content is better than tiny content.

26.9 Case B9: overlay-aware TikZ

```

\begin{tikzpicture}
\node[block] (a) {Input};
\node[block,right=of a] (b) {Model};
\uncover<2->{\node[block,right=of b] (c) {Output};}
\draw[->] (a)--(b);
\uncover<2->{\draw[->] (b)--(c);}
\end{tikzpicture}

```

Reserve nodes with `\uncover` when you want later overlays to keep the same layout. With `\only`, later additions may recenter the entire picture unless the bounding box is fixed.

Beamer failure checklist

Frame is not the same as output slide; overlay ranges matter; `<3>` is only slide 3 but `<3->` persists; `\only` removes space; code frames need `fragile`; columns must fit; crowded frames should be edited, not shrunk; and overlay commands in tables must be scoped so they do not hide following rows.

27 TikZ casebook: build diagrams from reusable primitives

Start with `\usepackage {tikz}`. Load only the libraries a diagram needs: commonly `positioning`, `arrows.meta`, `shapes.geometric`, `automata`, `calc`, `mindmap`, and `trees`. Plots, circuits, and schedules use the separate packages `pgfplots`, `circuitikz`, and `pgfgantt`.

27.1 Case Z1: coordinates, grid, axes, fill, and labels

```

\begin{tikzpicture}[scale=.8]
\draw[step=1cm,gray!30,thin] (-3,-3) grid (3,3);
\draw[->,thick] (-3.2,0)--(3.2,0) node[right]{$x$};
\draw[->,thick] (0,-3.2)--(0,3.2) node[above]{$y$};
\coordinate (0) at (0,0);
\draw[blue,thick] (0) circle (2);
\fill (0) circle (2pt) node[below left]{$0$};

```

```
\end{tikzpicture}
```

Every path normally ends with a semicolon. Coordinates may be literal or named. Options in brackets style the whole path.

27.2 Case Z2: reusable flowchart styles

```
\usetikzlibrary{positioning,arrows.meta,shapes.geometric}
\tikzset{
  startstop/.style={ellipse,draw,fill=red!10,minimum width=2.4cm},
  process/.style={rectangle,rounded corners,draw,fill=blue!10,
    minimum width=2.8cm,minimum height=9mm,align=center},
  decision/.style={diamond,draw,aspect=2,align=center,fill=yellow!15},
  flow/.style={-Stealth,thick}}
\begin{tikzpicture}[node distance=1.4cm]
\node[startstop] (s) {Start};
\node[process,below=of s] (p) {Read input};
\node[decision,below=of p] (d) {Valid?};
\node[process,right=of d] (fix) {Repair};
\node[startstop,below=of d] (e) {End};
\draw[flow] (s)--(p); \draw[flow] (p)--(d);
\draw[flow] (d)--node[right]{yes}(e);
\draw[flow] (d)--node[above]{no}(fix);
\draw[flow] (fix)--(p);
\end{tikzpicture}
```

|– and –| create orthogonal routes. Named nodes make later edits safer than hard-coded coordinates.

27.3 Case Z3: automaton with loops and bent edges

```
\usetikzlibrary{automata,positioning,arrows.meta}
\begin{tikzpicture}[node distance=2.8cm,on grid,auto,>=Stealth]
\node[state,initial] (q0) {$q_0$};
\node[state,accepting,right=of q0] (q1) {$q_1$};
\path[->]
(q0) edge[loop above] node{0} ()
     edge[bend left] node{1} (q1)
(q1) edge[loop above] node{1} ()
     edge[bend left] node{0} (q0);
\end{tikzpicture}
```

auto chooses label sides; explicit above, below, swap, and bends resolve collisions. Initial/accepting are node styles from the automata library.

27.4 Case Z4: a tree with level-specific spacing

```
\begin{tikzpicture}[
  every node/.style={circle,draw,minimum size=6mm},
  level 1/.style={sibling distance=3cm,level distance=1.5cm},
  level 2/.style={sibling distance=1.5cm}]
\node{Root}
  child{node{$v_1$}} child{node{$v_3$}} child{node{$v_4$}}
  child{node{$v_2$}} child{node{$v_5$}} child{node{$v_6$}};
\end{tikzpicture}
```

Trees are nested source structures. Control sibling distance per level rather than manually positioning every child.

27.5 Case Z5: loops for grids and repeated nodes

```
\begin{tikzpicture}
\foreach \x in {0,...,3}{
  \foreach \y in {0,...,3}{
    \draw (1.2*\x,1.2*\y) rectangle ++(1,1);
  }
}
\end{tikzpicture}
```

Loop variables can enter arithmetic coordinates. Braces around loop bodies prevent only the next command from being repeated accidentally.

27.6 Case Z6: a neural network generated by loops

```
\tikzset{neuron/.style={circle,draw,minimum size=7mm}}
\begin{tikzpicture}
\foreach \i in {0,1,2}\node[neuron] (I\i) at (0,{2-\i}) {};
\foreach \h in {0,1}\node[neuron] (H\h) at (2,{1.5-\h}) {};
\node[neuron] (O) at (4,1) {};
\foreach \i in {0,1,2}\foreach \h in {0,1}
  \draw[->] (I\i)--(H\h);
\foreach \h in {0,1}\draw[->] (H\h)--(O);
\end{tikzpicture}
```

Node names are constructed from loop variables. Three inputs times two hidden nodes produce six first-layer edges.

27.7 Case Z7: geometry with relative coordinates and calculation

```
\usetikzlibrary{calc}
\begin{tikzpicture}
\coordinate (A) at (0,0); \coordinate (B) at (4,0);
\coordinate (C) at (1.2,2.5);
\draw[thick,fill=green!8] (A)--(B)--(C)--cycle;
\coordinate (M) at ($(A)!.5!(B)$);
\draw[dashed] (C)--(M);
\node[below] at (M) {midpoint};
\end{tikzpicture}
```

The calc expression $(A)!.t!(B)$ interpolates from A to B. Relative coordinates $++(dx,dy)$ update the current point; $+(dx,dy)$ does not.

27.8 Case Z8: plotting a function or data

```
\usepackage{pgfplots}\pgfplotsset{compat=1.18}
\begin{tikzpicture}
\begin{axis}[width=8cm,height=5cm,xlabel=$t$,ylabel=$v_C$,
  grid=major,legend pos=south east]
\addplot[blue,thick,domain=0:5,samples=100]{1-exp(-x)};
\addlegendentry{model}
% \addplot table[x=t,y=v,col sep=comma]{data.csv};
\end{axis}
\end{tikzpicture}
```

Use many samples for a smooth nonlinear curve. A data file can be plotted by naming columns and separators; keep pgfplots syntax conceptually separate from core TikZ paths.

27.9 Case Z9: an electrical circuit

```
\usepackage[siunitx]{circuitikz}
\begin{circuitikz}
\draw (0,0) to[V,l=$v_{in}$] (0,3)
      to[R,l=$R$,i>^=$i$] (3,3)
      to[C,l=$C$,v^>=$v_C$] (3,0) -- (0,0);
\end{circuitikz}
```

Common component keys are R,C,L,V,I,D,switch,short; labels use l=, currents i=, and voltages v=. Arrow/label orientation modifiers may vary with package version, so recognize the structure rather than memorizing every arrow variant.

27.10 Case Z10: concise mind map and schedule recognition

```
\usetikzlibrary{mindmap,trees}
\begin{tikzpicture}[mindmap,grow cyclic,every node/.style=concept]
\node{CSE} child{node{Theory}} child{node{Systems}} child{node{AI}};
\end{tikzpicture}

\usepackage{pgfgantt}
\begin{ganttchart}[vgrid,hgrid]{1}{6}
\gantttitle{Weeks}{6}\n
\ganttbar{Design}{1}{2}\n \ganttbar{Build}{3}{5}
\end{ganttchart}
```

Mind maps use a TikZ library; Gantt charts use `pgfgantt`. These are plausible recognition questions but lower priority than nodes, paths, styles, loops, flowcharts, automata, plots, and circuits.

28 Recognition-only boundary: deliberately not expanded

The supplied references mention several powerful tools that are disproportionate for a short quiz. Know the name-purpose mapping, but do not spend memorization time on full syntax unless your teacher explicitly adds them:

- `physics` shorthand for derivatives and bra-ket notation; `mathtools` paired delimiters; `empheq` boxed equation systems.
- `mhchem/chemfig` chemistry; `forest/tikz-qtree` syntax trees; `bytefield` packet layouts; `qrcode` QR codes.
- `tabularray` all-in-one tables; `adjustbox` general scaling; `dcolumn` legacy decimal alignment.
- `pdfscape`, `pdfpages`, watermarks, margin notes, epigraphs, and todo notes.
- Glossaries, acronyms, indexes, `biblatex+biber`, counters/conditionals, and compiling subsets of very large multi-file books.

Practical exceptions retained in detail above are `longtable`, `siunitx` decimal columns, `threeparttable`, custom column types, `cleveref`, and `captionof`, because they directly solve common lab-quiz layouts.

Part VIII Practice Set

29 True/False: 60 questions

- | | |
|---|--|
| T1. The preamble lies inside the document environment. | T24. <code>\newtheorem {lemma} [theorem] {Lemma}</code> shares the theorem counter. |
| T2. Optional arguments normally use square brackets. | T25. <code>table</code> creates cells, while <code>tabular</code> supplies captions. |
| T3. A starred section is normally numbered. | T26. <code>p{3cm}</code> creates a fixed-width paragraph column. |
| T4. <code>\clearpage</code> flushes pending floats before continuing. | T27. <code>\multicolumn</code> requires the <code>multirow</code> package. |
| T5. A blank source line begins a new paragraph. | T28. Booktabs encourages vertical rules. |
| T6. <code>\emph</code> always produces italic text regardless of context. | T29. <code>\cline {2-3}</code> spans only columns 2 and 3. |
| T7. A size declaration can leak forward if it is not grouped. | T30. H float placement requires the <code>float</code> package. |
| T8. <code>xcolor</code> with option <code>table</code> supports table cell coloring. | T31. The exclamation mark in <code>[!htbp]</code> forces the float exactly here. |
| T9. <code>itemize</code> requires <code>enumitem</code> . | T32. <code>angle</code> is an <code>\includegraphics</code> option. |
| T10. <code>description</code> labels are supplied in brackets after <code>\item</code> . | T33. A blank line between subfigures can force a new row. |
| T11. A figure label should usually precede its caption. | T34. <code>wrapfigure</code> allows text beside a small image. |
| T12. Hyperlinks require <code>hyperref</code> . | T35. In a two-column article, <code>\linewidth</code> inside normal text is one column wide. |
| T13. Every mathematical character, even a single variable, belongs in math mode. | T36. <code>figure*</code> is used for a two-column-wide figure. |
| T14. <code>\$\$...\$\$</code> is preferred over <code>\[...\]</code> in LaTeX. | T37. <code>multicols</code> and a document-wide <code>twocolumn</code> option are identical mechanisms. |
| T15. <code>\sqrt [3]{x}</code> produces a cube root. | T38. <code>\citep</code> is a natbib command for parenthetical citations. |
| T16. <code>\mathbb {R}</code> normally needs an AMS symbol/font package. | T39. <code>\nocite {*}</code> prints only the cited database entries. |
| T17. <code>\sin(x)</code> and <code>\sin (x)</code> are typographically identical. | T40. <code>unsrt</code> orders numeric references by first citation. |
| T18. Every <code>\left</code> must have a matching <code>\right</code> in the same formula line. | T41. BibTeX normally needs only one LaTeX run. |
| T19. <code>\left .</code> can supply an invisible delimiter. | T42. <code>algorithm2e</code> and <code>algpseudocode</code> use exactly the same commands. |
| T20. <code>\cdots</code> is normally better than <code>\ldots</code> in a sum. | T43. <code>listings</code> can format program code without shell escape. |
| T21. The star in <code>align*</code> suppresses numbering. | T44. A Beamer frame always produces exactly one PDF page. |
| T22. <code>&</code> marks alignment points in <code>align</code> . | T45. <code>\pause</code> begins another overlay. |
| T23. <code>pmatrix</code> requires a column specification such as <code>cc</code> . | T46. <code><3-></code> means only overlay 3. |

- T47.** `\only` reserves space while hidden.
- T48.** `\uncover` reserves space for covered content.
- T49.** `\setbeamercovered {transparent}` can show covered items faintly.
- T50.** `\alert` is theme-aware emphasis.
- T51.** Beamer columns may safely total more than the text width.
- T52.** A frame containing `verbatim` normally needs `fragile`.
- T53.** `block`, `alertblock`, and `exampleblock` are Beamer structures.
- T54.** TikZ relative placement `right=of a` needs the positioning library.
- T55.** `\foreach` can generate repeated TikZ nodes or edges.
- T56.** TikZ `node[midway,above]` can label an edge.
- T57.** `pgfplots` uses an `axis` environment.
- T58.** `chemfig` is intended for circuit diagrams.
- T59.** `lipsum` generates placeholder text.
- T60.** Vector graphics are usually preferable for plots and diagrams.

30 Multiple choice: 50 questions

- M1.** Which is mandatory? (A) `\maketitle` (B) `\documentclass` (C) `\tableofcontents` (D) `\author`
- M2.** Which creates an unnumbered section? (A) `\section !` (B) `\section *` (C) `\section -` (D) `\nosection`
- M3.** Which flushes floats? (A) `\newpage` always (B) `\clearpage` (C) `\` (D) `\hfill`
- M4.** Which prints a literal percent sign? (A) `%` (B) `\%` (C) `%%` (D) `\percent`
- M5.** Which scopes a size change locally? (A) Braces (B) Brackets (C) Percent signs (D) Ampersands
- M6.** Which list is term-definition oriented? (A) `itemize` (B) `description` (C) `enumerate` (D) `tabular`
- M7.** Which package customizes list labels? (A) `enumitem` (B) `geometry` (C) `caption` (D) `float`
- M8.** Best label order for a figure? (A) label then caption (B) caption then label (C) either always (D) no caption
- M9.** Which prints the page of a label? (A) `\ref` (B) `\pageref` (C) `\cite` (D) `\url`
- M10.** Which prints custom hyperlink text? (A) `\url` (B) `\href` (C) `\ref` (D) `\footnote`
- M11.** Preferred unnumbered display syntax? (A) `$. . . $` (B) `\[. . . \]` (C) `\(. . . \)` (D) text mode
- M12.** Which gives $x_{i,j}$? (A) `xi,j` (B) `x_{i,j}` (C) `x[i,j]` (D) `x(i,j)`
- M13.** Which prints an upright sine operator? (A) `sin` (B) `\sin` (C) `\textit{sin}` (D) `\mathcal{sin}`
- M14.** Which denotes a set of reals? (A) `\mathcal{R}` (B) `\mathbb{R}` (C) `\textrm{R}` (D) `\vec{R}`
- M15.** Which supplies diagonal dots? (A) `\ldots` (B) `\vdots` (C) `\ddots` (D) `\cdots`
- M16.** Which is a one-sided fence trick? (A) `\left . . . \right |` (B) `\left . . .` (C) `\right . . .` (D) `| . . .`
- M17.** Which environment aligns at ampersands? (A) `verse` (B) `align` (C) `figure` (D) `itemize`
- M18.** Which matrix has square brackets? (A) `pmatrix` (B) `bmatrix` (C) `vmatrix` (D) `matrix`
- M19.** Which sets cases? (A) `cases` (B) `splitcase` (C) `ifmath` (D) `arraycase`
- M20.** Which defines a main operator in the preamble? (A) `\DeclareMathOperator` (B) `\newpage` (C) `\operator` (D) `\mathrm` only
- M21.** What actually builds a table grid? (A) `table` (B) `tabular` (C) `caption` (D) `arraystretch`
- M22.** Correct five-column spec: left, three centered, right? (A) `[lcccr]` (B) `{lcccr}` (C) `{|lcccr|}` (D) `{lcccr}`
- M23.** Which merges columns? (A) `\multirow` (B) `\multicolumn` (C) `\cmidrule` (D) `\rowcolors`
- M24.** Which is a booktabs partial rule? (A) `\cline` only (B) `\cmidrule` (C) `\hline` (D) `\vline`
- M25.** Which creates an auto-width text column? (A) `tabularx` and `X` (B) `table` and `T` (C) `array` and `A` (D) `longtable` and `L`
- M26.** Which table can break across pages? (A) `tabular` (B) `longtable` (C) `table` (D) `subtable`
- M27.** Which does not force exact placement? (A) `H` (B) `!` (C) neither (D) both
- M28.** Which prevents image distortion? (A) `clip` (B) `keepaspectratio` (C) `angle` (D) `scale=0`
- M29.** Which crops an image? (A) `trim=... ,clip` (B) `hfill` (C) `caption` (D) `ref`
- M30.** Which package supports subfigures here? (A) `subcaption` (B) `wrapfig` (C) `float` (D) `array`
- M31.** Which provides text wrapping beside a figure? (A) `figure*` (B) `wrapfigure` (C) `subfigure` (D) `axis`
- M32.** What spans two columns? (A) `figure` (B) `figure*` (C) `wrapfigure` (D) `minipage*`
- M33.** Which produces local columns? (A) `multicols` (B) `multirow` (C) `columns` in article (D) `\cmidrule`
- M34.** Which natbib command integrates author into prose? (A) `\citep` (B) `\citert` (C) `\nocite` (D) `\citeyearpar` only
- M35.** Which style follows citation order? (A) `plain` (B) `unsrt` (C) `abbrv` (D) `apalike`
- M36.** Which inserts all BibTeX entries? (A) `\cite {*}` (B) `\nocite {*}` (C) `\bibliography {*}` (D) `\ref {*}`
- M37.** Which pair implements the tutorial's Binary Search pseudocode? (A) `algorithm+algpseudocode` (B) `listings+xcolor` (C) `tikz+float` (D) `natbib+caption`

- M38.** Which environment holds formatted source code? (A) `algorithmic` (B) `lstlisting` (C) `versemath` (D) `codefloat`
- M39.** In Beamer, the source container is a: (A) slide (B) frame (C) pagebreak (D) block
- M40.** Which means overlay 2 onward? (A) `<2>` (B) `<2->` (C) `<-2>` (D) `<1-2>`
- M41.** Which does not reserve hidden space? (A) `\only` (B) `\uncover` (C) `\visible` (D) `\onslide +`
- M42.** Which produces theme-aware warning emphasis? (A) `\textit` (B) `\alert` (C) `\underline` (D) `\emph` only
- M43.** Which is a valid Beamer block? (A) `alertblock` (B) `warningtable` (C) `slideblock` (D) `framebox*`
- M44.** Which option is needed for a code-listing frame? (A) `fragile` (B) `twocolumn` (C) `draft` (D) `openright`
- M45.** Which library enables `right=of`? (A) `mindmap` (B) `positioning` (C) `calc` only (D) `trees`
- M46.** Which TikZ command repeats code? (A) `\loop` (B) `\foreach` (C) `\repeatnode` (D) `\multido` in core TikZ
- M47.** Which package draws function plots? (A) `pgfplots` (B) `pgfgantt` (C) `skak` (D) `verse`
- M48.** Which package draws circuits? (A) `chemfig` (B) `circuitikz` (C) `tikz-qtree` (D) `musixtex`
- M49.** Which package makes Gantt charts? (A) `pgfgantt` (B) `pgfplots` (C) `booktabs` (D) `lettrine`
- M50.** Which command makes a drop capital? (A) `\lettrine` (B) `\Huge` (C) `\capital` (D) `\dropcap` in the tutorial

31 Output prediction and short questions

31.1 Predict or explain

- P1.** What changes when the star is removed from `align*`?
- P2.** Explain the output difference between `\textit {X}` and nested `\emph {X}`.
- P3.** What does `\begin {tabular}{|l|c|r|}` specify?
- P4.** Why can `[h]` still place a figure elsewhere?
- P5.** What happens when two `.50\linewidth` subfigures also have interword space between them?
- P6.** In a two-column article, contrast image width `\linewidth` inside `figure` with `\textwidth` inside `figure*`.
- P7.** What does `\nocite {\lmpart1994latex}` print at its location?
- P8.** How many PDF overlays are generated by `A\pause` `B\pause` `C` in one frame?
- P9.** Contrast `\only <2>{X}` and `\uncover <2>{X}`.
- P10.** What nodes and edges are generated by two nested `\foreach` loops over 3 inputs and 2 hidden nodes?

31.2 Short answers

- S1.** Distinguish preamble and body, with two examples from each.
- S2.** Explain mandatory arguments, optional arguments, declarations, and environments.
- S3.** List the math modes and state when each is appropriate.
- S4.** Explain automatic versus manual delimiter sizing and a one-sided delimiter.
- S5.** Compare `equation`, `align`, `multline`, `cases`, and matrix environments.
- S6.** Distinguish `table` from `tabular`; then explain four column specifications.
- S7.** Compare `\hline`, `\cline`, and the `booktabs` rule commands.
- S8.** Explain three ways to merge/structure complex table cells.
- S9.** Explain all six float-placement characters covered in the tutorial.
- S10.** Give syntax for a single figure, two independent images, two subfigures, a wrapped figure, and a two-column-wide figure.
- S11.** Explain why `\linewidth` is often safer than `\textwidth`.
- S12.** Describe the complete BibTeX/natbib workflow and compile order.
- S13.** Contrast plain `\cite`, `\citet`, `\citep`, and `\nocite`.
- S14.** Contrast the two pseudocode package families in the sources.
- S15.** Distinguish a Beamer frame from a slide/overlay.
- S16.** Explain `\pause`, overlay ranges, `\only`, `\uncover`, and transparent covered content.
- S17.** Write the rules for usable Beamer columns and staged tables.
- S18.** Explain TikZ nodes, named coordinates, styles, relative placement, and paths.
- S19.** Explain how nested `\foreach` loops build a neural-network diagram.
- S20.** Match five specialized packages to their purpose.

32 Code-writing tasks

Write complete minimal fragments, including package dependencies where relevant.

- W1.** An article with title, author, abstract, contents, two sections, and a page reference to Section 2.
- W2.** A nested list with Roman outer labels, alphabetic inner labels, and one custom arrow bullet.
- W3.** A numbered aligned derivation with one unnumbered intermediate line.
- W4.** A piecewise function and a 3-by-3 bracketed matrix.
- W5.** A theorem and lemma sharing a counter, with a labelled lemma and proof.
- W6.** A professional table with four columns, a heading spanning the first three, one cell spanning two rows, alternating row color, caption, and label.
- W7.** A figure cropped with `trim`, constrained to width and height without distortion, captioned and referenced.
- W8.** A 2-by-2 subfigure grid with subcaptions and a common caption.
- W9.** A paragraph with a right-wrapped image.
- W10.** A two-column article containing a one-column figure and a full-width three-image figure.
- W11.** A `.bib` article entry, a textual natbib citation, a parenthetical citation, and bibliography commands.
- W12.** A numbered Binary Search algorithm using `algorithmic` commands.
- W13.** A Beamer title frame, outline, and a frame whose four bullets appear one at a time.
- W14.** A Beamer frame with 40/60 text-image columns and an alert block.
- W15.** A fragile Beamer frame containing a Python listing.
- W16.** A TikZ flow Input $\rightarrow$ Process $\rightarrow$ Output using a reusable block style.
- W17.** A TikZ 4-by-4 grid created with nested loops.
- W18.** A TikZ automaton with an initial state, accepting state, loop, and bent transition.

33 Debugging and correction tasks

Find every error or weakness and repair it.

- D1.** `\documentclass [article]{11pt}`
- D2.** `\includegraphics [0.5\linewidth ]{a.png}`
- D3.** `\begin {tabular}[lccr] A & B & C & D \end {tabular}`
- D4.** `xij2` when the intended subscript is ij .
- D5.** `\left (\frac {a}{b})`
- D6.** `\begin {align*}a=b &+c \\ d=e &+f\end {align*}` when equals signs should align.
- D7.** Figure source has `\label {fig:x}` before `\caption {X}`.
- D8.** `\begin {figure}[T]` in an article.
- D9.** Two `0.50\linewidth` subfigures separated by a blank line.
- D10.** A `longtable` placed inside a `table` float.
- D11.** `\citep {key}` without loading `natbib`.
- D12.** A bibliography run sequence of LaTeX only once.
- D13.** `\begin {frame}{Code}\begin {verbatim}...\end {verbatim}\end {frame}`.
- D14.** Beamer columns of `.6\textwidth` and `.6\textwidth` .
- D15.** `\item <3>` when the item should remain visible from overlay 3 onward.
- D16.** TikZ `right=of A` without loading `positioning`.

34 Advanced case-based challenge set

These questions target the variations added from the practice packs and the parallel Beamer/TikZ casebooks.

- A1.** Choose and justify the best environment for four aligned lines that must share exactly one equation number.
- A2.** Write three aligned lines in which only the final line is numbered and labelled.
- A3.** When is `alignat` preferable to `align`, and what does its mandatory number mean?

- A4.** Create three related equations numbered (5a)–(5c) and give the group one reference key.
- A5.** Typeset a derivative evaluated from 0 to 1 with a correctly growing vertical bar.
- A6.** Typeset a tensor with a leading upper index and leading lower index without inserting a visible base symbol.
- A7.** Give one legitimate use and one danger of `\tag`.
- A8.** Design a two-tier table header: “Method” spans two rows; “Score” spans Accuracy and F1.
- A9.** A header row must be dark blue with white text. In what order should `\rowcolor` and the cell contents appear?
- A10.** Make a full-width three-column table whose final two columns wrap text automatically.
- A11.** Put a figure and an independently numbered table side by side without nesting two floats.
- A12.** Explain why `threeparttable` is preferable to an ordinary footnote placed directly inside `tabular`.
- A13.** Which package and column type align numbers at the decimal marker? Give a two-column skeleton.
- A14.** Place two images side by side under one common caption, with no subfigure letters.
- A15.** Crop 10pt from the left and 20pt from the top of an image; explain why one extra option is essential.
- A16.** Place a footnote marker in a float and its text outside the float.
- A17.** Match `\eqref`, `\autoref`, `\nameref`, and `\cref` to their characteristic output.
- A18.** Define a reusable Python listings style, then show the command that imports an external file with that style.
- A19.** Continue numbering in a second list and create an inline enumerated list; identify the relevant package option.
- A20.** Contrast Beamer’s `\only`, `\uncover`, and `\visible` in terms of visibility and reserved space.
- A21.** A staged Beamer table jumps horizontally as columns appear. Repair the design without showing future values.
- A22.** With TikZ’s `calc` library, name the point halfway between nodes A and B and draw a node there.
- A23.** A network has 3 input, 4 hidden, and 2 output nodes, fully connected only between adjacent layers. How many edges should nested loops create?
- A24.** A TikZ diagram changes position between overlays when a node is introduced with `\only`. Give two stable repairs.

Part IX

Answer Key

35 True/False answers

T1. F	T21. T	T41. F
T2. T	T22. T	T42. F
T3. F	T23. F	T43. T
T4. T	T24. T	T44. F
T5. T	T25. F	T45. T
T6. F	T26. T	T46. F
T7. T	T27. F	T47. F
T8. T	T28. F	T48. T
T9. F	T29. T	T49. T
T10. T	T30. T	T50. T
T11. F	T31. F	T51. F
T12. T	T32. T	T52. T
T13. T	T33. T	T53. T
T14. F	T34. T	T54. T
T15. T	T35. T	T55. T
T16. T	T36. T	T56. T
T17. F	T37. F	T57. T
T18. T	T38. T	T58. F
T19. T	T39. F	T59. T
T20. T	T40. T	T60. T

36 MCQ answers

M1. B	M11. B	M21. B	M31. B	M41. A
M2. B	M12. B	M22. B	M32. B	M42. B
M3. B	M13. B	M23. B	M33. A	M43. A
M4. B	M14. B	M24. B	M34. B	M44. A
M5. A	M15. C	M25. A	M35. B	M45. B
M6. B	M16. A	M26. B	M36. B	M46. B
M7. A	M17. B	M27. B	M37. A	M47. A
M8. B	M18. B	M28. B	M38. B	M48. B
M9. B	M19. A	M29. A	M39. B	M49. A
M10. B	M20. A	M30. A	M40. B	M50. A

37 Prediction and short-answer marking points

Prediction

- P1. Lines receive equation numbers (subject to suppression commands).
- P2. `\textit` mechanically italicizes; `\emph` changes emphasis relative to context and may become upright inside emphasis.
- P3. Three columns: left, centered, right, with vertical rules at both edges and between columns.
- P4. `h` is permission/preference under the float algorithm, not a command.
- P5. Their widths plus spacing exceed the line; the second usually moves down or creates overflow output.
- P6. In ordinary `\figure`, `\linewidth` is one column; in `\figure*`, the useful full span is approximately `\textwidth`.

- P7.** Nothing locally; it marks the entry for inclusion in the bibliography.
- P8.** Three overlays/pages.
- P9.** `only` removes X and its space when inactive; `uncover` preserves the space.
- P10.** Six connections from every input to every hidden node; if followed by two hidden-to-output edges, eight total.

Short-answer essentials

Use Sections 2–20 as model answers. Full credit requires package dependencies, correct brace/bracket syntax, and the following contrasts: preamble/body; inline/display/numbered math; automatic/manual fences; floating wrapper/grid; local/current width/full text width; frame/overlay; `only/uncover`; algorithm ecosystem/listing; TikZ style/node/path/loop.

38 Code-writing reference solutions

The tasks intentionally permit several correct implementations. These compact patterns show the required core.

- W1.** Use the skeleton in Section 2; add `abstract`, `\tableofcontents`, `\section {A}`, `\section {B}\label {sec:b}`, and `page~\pageref {sec:b}`.
- W2.** Use `enumitem`: outer `enumerate[label=(\roman *)]`, inner `enumerate[label=(\alph *)]`, and `\item [ $to $]`.
- W3.** Use `align`; place `&` before each equals sign and `\nonumber` on the intermediate row.
- W4.** Use `cases` and `bmatrix`; separate cells by `&` and rows by `\\`.
- W5.** Load `amsthm`; use the two `\newtheorem` lines from Section 9 and a `proof` environment.
- W6.** Load `booktabs`, `multirow` and `[table]{xcolor}`; use `\multicolumn {3}{c}`, `\multirow {2}{*}`, `\rowcolors`, then caption and label.
- W7.** Load `graphicx`; use all requested options, for example:

```
\includegraphics[width=.8\linewidth,height=6cm,keepaspectratio,
trim=1cm 1cm 1cm 1cm,clip]{image.png}
```

- W8.** Repeat the two-subfigure pattern twice; explicitly start a second row; keep each width around `.48\linewidth`.
- W9.** Load `wrapfig`; use `\begin {wrapfigure}{r}{.35\linewidth }`, image width `\linewidth`, caption, label, then sufficient prose.
- W10.** Select `twocolumn`; ordinary `figure` for one column and `figure*[t]` with three `.31\textwidth` subfigures.
- W11.** Use an `@article` or `@inproceedings` entry; load `natbib`; use `\citete`, `\citep`, `style`, and `database` commands.
- W12.** Follow the Section 16 `algorithm+algorithmic` pattern and include `[1]` for line numbering.
- W13.** Use Beamer skeleton plus `\begin {itemize}[<+>]` with four items.
- W14.** Use `columns[onlytextwidth]` with widths near `.38` and `.58`; put image at `\linewidth`; add `alertblock`.
- W15.** Add `[fragile]` to the frame and a `lstlisting`; load `listings`.
- W16.** Define `block` and `line` styles; create three named nodes with `right=of`; draw two arrows.
- W17.** Nest `\foreach \x in {0,...,3}` and `\foreach \y in {0,...,3}`, then draw a rectangle at a coordinate depending on both.
- W18.** Load `automata`, `positioning`; create `state`, `initial` and `state,accepting` nodes and draw loop/bent arrow paths.

39 Debugging solutions

- D1.** `\documentclass [11pt]{article}`: class is mandatory brace argument; point size is optional.
- D2.** `\includegraphics [width=.5\linewidth ]{a.png}`.
- D3.** `\begin {tabular}{lccr} A & B & C & D\\ \end {tabular}`: mandatory column spec and row terminator.
- D4.** `x_{ij}^2`.
- D5.** `\left (\frac {a}{b}\right )`.
- D6.** Put `&` before both equals signs: `a&=b+c` and `d&=e+f`.
- D7.** Move label after caption.
- D8.** Replace T with a valid combination such as `t` or `!htbp`; capital H is the special float-package option.
- D9.** Reduce widths (e.g. `.48\linewidth`), remove the blank paragraph, and separate with `\hfill`.
- D10.** Remove the `table` wrapper; `longtable` manages its own page-breaking caption structure.

- D11. Load `natbib` or use built-in `\cite` with plain BibTeX.
- D12. Run LaTeX, BibTeX, LaTeX, LaTeX; also verify keys/database/style.
- D13. Add `[fragile]` to the frame.
- D14. Reduce widths so their sum plus gaps fits, e.g. `.38` and `.58`.
- D15. Use `\item <3->`.
- D16. Add `\usetikzlibrary {positioning}` or place with explicit coordinates.

40 Advanced challenge blueprints

- A1. Put `aligned` inside `equation`; the outer environment supplies the single number.
- A2. Use `align`, add `\notag` (or `\nonumber`) to the first two rows, and place `\label` on the last row.
- A3. Use `alignat` when several explicitly controlled alignment pairs need no automatic inter-pair spacing; the argument is the number of pairs.
- A4. Wrap three labelled equations in `subequations`; label the outer environment for the parent number and individual rows for the letters.
- A5. Use `\left . expression \right |_{0}^{1}`; the invisible left delimiter balances the growing right bar.
- A6. Use an empty group as the base: `{ }^{\{a\}_{b}}T`; add `\vphantom {T}` inside the group if vertical alignment needs correction.
- A7. `\tag {R}` deliberately replaces an automatic displayed-equation tag. Repeated or arbitrary tags can destroy consistent numbering and references.
- A8. Use a two-row `tabular`: `\multirow {2}{*}{Method}` and `\multicolumn {2}{c}{Score}`, followed by `\cline {2-3}` and the two subheadings.
- A9. Put `\rowcolor {blue!70}` before the first cell; then apply white text to the header cells.
- A10. Load `tabularx` and use `\begin {tabularx}{\textwidth }{lXX}`. The two X columns share the remaining width.
- A11. Use two adjacent `minipages`. Put the image and `\captionof {figure}` in one, and the `tabular` plus `\captionof {table}` in the other; load `caption`.
- A12. Notes belong below the table, not inside alignment cells. `threeparttable` connects `\tnote` markers to compact `tablenotes` without float-footnote problems.
- A13. Load `siunitx`; use a skeleton such as `\begin {tabular}{l S[table-format=3.2]}`.
- A14. Use one `figure` containing two adjacent `minipages` (or two images separated by `\hfill`) and only one outer `\caption`.
- A15. Use `\includegraphics [trim=10pt 0 0 20pt,clip]{file}`. The order is left–bottom–right–top; `clip` actually removes the trimmed region.
- A16. Put `\footnotemark` at the required place inside the float and `\footnotetext {...}` immediately after the float.
- A17. `\eqref` gives a parenthesized equation number; `\autoref` includes the object type; `\nameref` prints the target title; `\cref` intelligently formats types, lists, and ranges.
- A18. Define the style with `\lstdefinestyle {py}{...}`, then use `\lstinputlisting [style=py]{file.py}`.
- A19. Load `enumitem` with option `inline`; use `[resume]` on the second list and `enumerate*` for the inline list.
- A20. `\only` removes absent material and its space. `\uncover` and `\visible` reserve the space; their covered-state treatment differs with Beamer settings.
- A21. Keep the cells present and hide future contents with `\phantom`, or use fixed-width columns plus overlay commands that reserve space.
- A22. Load `calc`; use `\coordinate (M) at ($(A)!0.5!(B)$)`; and place or draw the node at (M).
- A23. $3 \times 4 + 4 \times 2 = 20$ edges.
- A24. Use `\uncover` so space is reserved, or establish a fixed bounding box with `\useasboundingbox`; fixed coordinates are also helpful.

41 Final rapid-recall tables

Dependencies to memorize

Images: <code>graphicx</code>	Color/table color: <code>xcolor[table]</code>	Pseudocode A: <code>algorithm + algpseudocode</code>
Exact H: <code>float</code>	Math: <code>amsmath,amssymb</code>	Pseudocode B: <code>algorithm2e</code>
Wrapped image: <code>wrapfig</code>	Theorems: <code>amsthm</code>	Local columns: <code>multicol</code>
Subfigures: <code>subcaption</code>	Custom lists: <code>enumitem</code>	Drawing: <code>tikz</code>
Row merge: <code>multirow</code>	Links: <code>hyperref</code>	Plots: <code>pgfplots</code>
Professional rules: <code>booktabs</code>	Citations: <code>natbib</code>	
Auto-width table: <code>tabularx</code>	Code: <code>listings</code>	

Syntax traps

Braces are mandatory arguments; brackets are options. Put labels after captions. `!` relaxes float constraints; `H` forces here. `\linewidth` follows the current container. No blank paragraph between same-row subfigures. `\only` removes space; `\uncover` preserves it. Frame count and slide count can differ. Listings in Beamer need a fragile frame. Run BibTeX in the correct compilation cycle.